

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** CE

Commented [U1]: Please enter the name in Arial 12 font

2. **Subject Name:** Structural Design Sessional **L-T-P:** 0-0-6 **Credits:**4

Commented [U2]: Please enter the details (use Arial 12 font)

3. **Pre-requisites:** 1)DESIGN OF RC STRUCTURES. 2)STRUCTURAL ANALYSIS

Commented [U3]: Please enter the pre-requisites (subject name and subject no. in Arial 12 font)

4. **Syllabus and reference books:**

Commented [U4]: 1.Please type the syllabus and reference books in the box provided below.

Syllabus: Architectural and structural plan and elevation, Walkthrough using software, Manual calculation of various loads, analysis and design, Analysis and design of structure using software, Comparison of results with the manual computation, Structural design and detailing of beams, columns, staircase and foundation, Design of shear wall, Quantity estimation, Design of water distribution systems and road network, Report generation.

Please use "Arial" font with a font size of "12" and a line spacing of 1.0 for syllabus and reference books. Please use Justified Text (Ctrl+J) for the content entered in the boxes
Reference books should be numbered as 1), 2), 3)
Please provide all bibliographic details for the reference books

Reference Books:

- 1) Reinforced Concrete Design of Tall Buildings by B. S. Taranath; CRC Press
- 2) Structural Design of Multistoreyed Building by U. H. Vary; South Asian Publishers
- 3) Analysis and Design of Substructures by S. Saran; Oxford & IBH
- 4) Theory and Practice of Foundation Design by N. N. Som & S. C. Das; Printice-Hall of India
- 5) Estimating, costing, specification and valuation in civil engineering by M. Chakraborti,

5. **Lecture-wise break-up:**

Commented [U5]: Please provide lecture-wise break-up in the table given below:

Sl. No.	Topic	No. of lectures
1.	Planning: Design basis, Site layout plan, Architectural plans of the multi-storey buildings, elevations and sections, Walkthrough using software	12
2.	Analysis and design - Manual: Calculation of gravity, live, wind and earthquake loads as per BIS Codes, Sample hand calculation of a 2D frame for the gravity and lateral loads, Sample manual design calculation for beam, column, slab and footing, Structural detailing of beam, column, slab and footing, Detailing of junction, ductile detailing, Design of shear wall	30
3.	Analysis and design - Software: 3D Modeling of building, Load generation, Analysis of 3D frame,	12
4.	Design of multi-storey buildings considering the	18

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	appropriate load combination, design of beams, columns, foundations, staircase, shear walls.	
5.	Reinforcement detailing and quantity estimation: Reinforcement drawings, water tanks and water distribution network, road network, quantity estimation, and report generation.	
Total number of hours		72

Commented [U7]: Please convert the template into PDF and upload it in ERP